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GEOGRAPHY · The Earth in the Solar System

UNDERSTAND IT FIRST

What is it? Geography starts with **our home in space**. The **Earth** is one of **eight planets** that go around the **Sun**; the Sun is one of billions of stars in our **galaxy, the Milky Way**. The Earth's **shape, its spin (rotation) and its journey round the Sun (revolution)**, together with the **tilt of its axis**, explain **day and night, seasons, time zones, the length of days and eclipses**. Exams ask both **facts** (which planet is hottest?) and **reasoning** (why are there seasons? what time is it at a given longitude?). Learn it as **universe → Sun → planets → small bodies → Earth's shape and motions → latitudes and longitudes → Moon and eclipses**.

1. Universe, galaxy and the Sun

- **Universe:** began with the **Big Bang** about **13.8 billion years ago** (Georges Lemaître; Edwin Hubble showed the universe is **expanding**). **Galaxy:** a huge group of stars, gas and dust; ours is the **Milky Way (Akash Ganga)**, a **spiral** galaxy; our nearest big neighbour is **Andromeda**. **Light year** = distance light travels in a year, about **9.46 trillion km** (a unit of **distance**, not time). **Astronomical Unit (AU)** = average Earth–Sun distance, about **150 million km**.
- **Solar System:** formed about **4.6 billion years ago** from a spinning cloud of gas and dust (**nebular hypothesis**, Kant and Laplace). **The Sun** holds about **99.8% of its mass**; it is a **medium-sized star** made mostly of **hydrogen and helium**; it shines by **nuclear fusion** (hydrogen → helium). Surface (**photosphere**) about **5,500°C**, core about **15 million °C**; outer layers **chromosphere** and **corona** (seen in a total solar eclipse). **Sunspots** are cooler, darker patches that follow an **11-year cycle**. Sunlight reaches Earth in about **8 minutes 20 seconds**. India studies the Sun with **Aditya-L1** (Day 19).

2. The planets and other bodies

The eight planets

Planet	Remember
Mercury	**Smallest**, **closest** to the Sun, **fastest** orbit (88 days); **no moon**; huge day–night temperature range.
Venus	**Hottest** planet (thick **CO_2** atmosphere, greenhouse effect); **brightest** in our sky (**morning and evening star**); **Earth's twin** in size; rotates **east to west** (**retrograde**); a day longer than its year; **no moon**.
Earth	**Densest** planet; the **'Blue Planet'** (about **71% water**); only known planet with **life**; **one** moon.
Mars	**'Red Planet'** (iron oxide dust); **Olympus Mons**, the highest volcano in the Solar System; moons **Phobos and Deimos**; India's **Mangalyaan (2014)**.

Jupiter	**Largest**; **fastest spin** (about 10 hours); **Great Red Spot** (a giant storm); **Ganymede**, the **largest moon** in the Solar System.
Saturn	Bright **rings** of ice and rock; **least dense** (would float on water); **Titan** has a thick atmosphere; has the **most known moons**.
Uranus	Axis tilted about **98°**, so it **spins on its side**; also rotates **east to west**; blue-green from **methane**; discovered by **William Herschel (1781)**.
Neptune	**Farthest** planet; **strongest winds**; found by **mathematical prediction (1846)**; moon **Triton** orbits backwards.

- **Inner planets** (Mercury to Mars) are small and **rocky** (**terrestrial**); **outer planets** (Jupiter to Neptune) are huge **gas and ice giants** (**Jovian**) with rings and many moons. **Pluto** was reclassified as a **dwarf planet** by the **IAU in 2006** (it has not 'cleared its orbit'); other dwarf planets: **Ceres, Eris, Haumea, Makemake**.
- **Small bodies: asteroids** (rocky pieces, mostly in the **asteroid belt between Mars and Jupiter**; Ceres is the largest); **comets** (ice and dust; a tail that **points away from the Sun**; **Halley's comet** returns every **76 years**, next in **2061**); **meteors** ('shooting stars' that burn up in the air) and **meteorites** (pieces that reach the ground, e.g. **Lonar crater**, Maharashtra); **Kuiper belt** beyond Neptune and the distant **Oort cloud** (home of long-period comets).

3. The Earth: shape and motions

Earth's revolution and the seasons

Point	Details
Shape and size	Not a perfect sphere: slightly **flattened at the poles** and bulging at the equator, called a **geoid** (oblate spheroid). Equatorial diameter about **12,756 km**, polar about **12,714 km**; circumference about **40,075 km**. Third planet from the Sun, about **150 million km** away.
Rotation	Spins **west to east** on its axis in about **24 hours** (23 h 56 min against the stars). Causes **day and night**, the Sun's apparent rise in the **east**, and the **Coriolis force** that bends winds and currents. The boundary between day and night is the **circle of illumination**.
Revolution	Goes round the Sun in about **365¼ days** in an **elliptical** orbit; the extra quarter day gives a **leap year** every four years. **Perihelion** (closest, about **147 million km**) around **3 January**; **aphelion** (farthest, about **152 million km**) around **4 July**.
Seasons	Caused by the **23.5° tilt** of the axis (it makes **66.5°** with the orbital plane), which **always points the same way**, not by distance from the Sun. **21 June**: summer solstice, Sun overhead at the **Tropic of Cancer**, longest day in the north. **22 December**: winter solstice, Sun over the **Tropic of Capricorn**. **21 March and 23 September**: **equinoxes**, Sun over the **Equator**, day and night nearly **equal everywhere**.

4. Latitudes, longitudes and time

- **Latitudes** are imaginary circles **parallel to the Equator (0°)**, measured **north or south** up to **90°** at the poles. Key ones: **Tropic of Cancer (23.5° N)**, **Tropic of Capricorn (23.5° S)**, **Arctic Circle (66.5° N)**, **Antarctic Circle (66.5° S)**. The Tropic of Cancer passes through **eight Indian States**: Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura and Mizoram. Heat zones: **torrid** (between the tropics), **temperate**, **frigid**.
- **Longitudes (meridians)** run **pole to pole**; **0°** is the **Prime Meridian at Greenwich** (London), up to **180° east and west**. The Earth turns **360° in 24 hours**, so **15° = 1 hour** and **1° = 4 minutes**; places to the **east are ahead** in time. **IST** is based on **82.5° E** (near **Mirzapur**, UP), so it is **5 h 30 min ahead of GMT**; this meridian crosses **UP, MP, Chhattisgarh, Odisha and Andhra Pradesh**. India spans about **30° of longitude**, so the Sun rises about **two hours earlier** in Arunachal Pradesh than in Gujarat. The **International Date Line** roughly follows **180°**; crossing it eastward you **repeat a day**, westward you **skip a day**.

5. The Moon and eclipses

- **The Moon** is about **384,400 km** away and about **one-quarter** of Earth's diameter. It goes round the Earth in about **27.3 days**, but new moon to new moon takes about **29.5 days**. It **rotates once in the same time as it revolves**, so we always see the **same face**. Its gravity (with the Sun's) causes **tides**: **spring tides** at new and full moon, **neap tides** at quarter moons.
- **Solar eclipse:** the **Moon comes between the Sun and the Earth**; only on a **new moon** day. **Lunar eclipse:** the **Earth comes between the Sun and the Moon**; only on a **full moon** night. Eclipses do **not happen every month** because the Moon's orbit is **tilted about 5°** to the Earth's orbit. An **annular** solar eclipse leaves a **ring of fire** when the Moon is farther away.

EXAM TRAP

Big Bang 13.8 bn yrs; **Milky Way** spiral; **light year 9.46 trillion km** (distance); **AU ~150 million km**; Solar System **4.6 bn yrs**; Sun **99.8%** mass, **fusion**, surface **~5,500°C**, core **~15 million °C**, light **8 min 20 s**, sunspots **11 yrs**; planets **MVEMJSUN**: Mercury smallest, no moon; **Venus hottest, retrograde**; Earth **densest**; Mars **Olympus Mons, Phobos, Deimos**; Jupiter **largest, Ganymede**; Saturn **least dense, rings**; Uranus **98° tilt**; Neptune **farthest, windiest**; **Pluto dwarf (IAU 2006)**; asteroid belt **Mars–Jupiter**; **Halley 76 yrs (2061)**; geoid; **12,756 / 12,714 km**; rotation **W→E**; **365¼**; perihelion **3 Jan**, aphelion **4 July**; tilt **23.5°**; solstices **21 June, 22 Dec**; equinoxes **21 Mar, 23 Sept**; Tropic of Cancer **8 States**; **15° = 1 h**; **IST 82.5° E (5 States incl. AP), +5:30**; IDL **180°**; Moon **384,400 km**, **27.3 / 29.5 days**; solar eclipse **new moon**, lunar **full moon**; **5° tilt**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. Venus rotates from east to west.

2. Mercury has no natural satellite.

3. Jupiter is the least dense planet.

4. Uranus rotates almost on its side.

Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2 and 4 only

D. 1, 2, 3 and 4

Answer: C. Saturn, not Jupiter, is the least dense planet.

PRACTICE

Match List-I (position of the Earth) with List-II (approximate date):

List-I (position of the Earth)	List-II (approximate date)
A. Perihelion	1. 23 September
B. Aphelion	2. 21 June
C. Summer solstice (Northern Hemisphere)	3. 4 July
D. Autumnal equinox	4. 3 January

(1) A → 4 B → 3 C → 2 D → 1

(2) A → 3 B → 4 C → 2 D → 1

(3) A → 4 B → 2 C → 3 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (1) Perihelion early January; aphelion early July; solstice 21 June; equinox 23 September.

PRACTICE

Which one of the following statements is NOT correct?

A. Sunlight takes a little over eight minutes to reach the Earth.

B. The Earth is the densest planet in the Solar System.

C. Pluto was reclassified as a dwarf planet in 2006.

D. A lunar eclipse can occur only on a new moon day.

Answer: D. A lunar eclipse occurs on a full moon night; a solar eclipse occurs on a new moon day.

PRACTICE

Assertion (A): Places on the Equator have days and nights of nearly equal length throughout the year.
Reason (R): The Earth's axis is tilted at about 23.5° from the perpendicular to its orbital plane.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: B. Both are true. Equal days at the Equator happen because the circle of illumination always cuts the Equator in half; the tilt explains seasons and unequal days elsewhere.

PRACTICE

When it is 12 noon at Greenwich (0°), what is the local time at a place on 82.5° E longitude?

A. 6:30 a.m.

B. 5:30 p.m.

C. 6:30 p.m.

D. 4:30 p.m.

Answer: B. 82.5 × 4 minutes = 330 minutes = 5 h 30 min ahead (east), so 5:30 p.m.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the universe, galaxy and Sun facts. 2. Write the eight planets with one fact each, then dwarf planets and small bodies. 3. Write Earth's shape and motions with dates, latitudes, longitude–time rules and IST, and the Moon and eclipses.

POLITY · Balwant Rai Mehta and Ashok Mehta Committees; the 73rd Amendment (Panchayati Raj)

UNDERSTAND IT FIRST

What is it? Panchayati Raj means **rural local self-government**: villagers elect their own bodies to plan and run **local development**. The **Community Development Programme (1952)** failed because people had **no say** (Day 24). So committees recommended **democratic decentralisation**. States set up panchayats, but they were often **not elected on time**, had **little money** and **no power**. The **73rd Amendment (1992)** gave them **constitutional status**, making elections, reservation and finance **compulsory**. Learn it as **Balwant Rai Mehta → Ashok Mehta → later committees → 73rd Amendment features → exemptions and PESA → AP's system**.

The road to the 73rd Amendment

1. The committees

Committee	Recommendations
Balwant Rai Mehta (appointed January 1957; report November 1957)	Studied the **CDP and NES**; recommended **'democratic decentralisation'** through a **three-tier** system: **gram panchayat** (village), **panchayat samiti** (block) and **zila parishad** (district). Village panchayat **directly elected**; samiti and zila parishad **indirectly elected**. The **panchayat samiti** to be the **executive** body; the **zila parishad** an **advisory and supervisory** body with the **Collector as chairman**; real **powers and resources** to be transferred. The **National Development Council** accepted it in **January 1958**. **Rajasthan** was first (**Nagaur, 2 October 1959**, by Nehru); **Andhra Pradesh** followed in **1959**.
Ashok Mehta (appointed December 1977 by the Janata Government; report August 1978)	Recommended a **two-tier** system: **zila parishad** (district) and **mandal panchayat** (a group of villages, about **15,000–20,000** people). The **zila parishad** to be the **executive** body and the main unit of planning below the State. **Political parties** may take part officially; compulsory **taxation powers**; **reservation for SCs and STs**; **nyaya panchayats** separate from development panchayats; a **minister for Panchayati Raj** in each State; **social audit**. Not implemented nationally because the **Janata Government fell**, but **Karnataka, West Bengal and Andhra Pradesh** used its ideas (AP's **mandals**, 1985–86).
Later committees	**G.V.K. Rao (1985)**: development had been **bureaucratised**, making Panchayati Raj like **'grass without roots'**; the **district** should be the key unit. **L.M. Singhvi (1986)**: give panchayats **constitutional recognition**, regular elections and **nyaya panchayats**. **P.K. Thungon (1988)** also backed constitutional status. The **64th Amendment Bill (1989)** passed the Lok Sabha but **failed in the Rajya Sabha**.

2. The 73rd Constitutional Amendment Act, 1992

Passed in **December 1992** (Narasimha Rao Government), it came into force on **24 April 1993**, now celebrated as **National Panchayati Raj Day**. It added **Part IX** (**Articles 243 to 243-O**) and the **Eleventh Schedule** (**29 subjects**). It turned **Art 40** (a Directive Principle) into a working reality. Some provisions are **compulsory** for States; others are left to the **State's choice**.

Compulsory provisions	Left to the States (voluntary)
Gram Sabha of all voters in the village (**Art 243A**).	**Powers and functions** of the Gram Sabha.
Three tiers: village, intermediate and district (**243B**); States with population **below 20 lakh** may **skip the intermediate tier**.	How the **village chairperson** (sarpanch) is elected; **voting rights** for MPs and MLAs in panchayats.
Direct election of members at all levels; **indirect election** of chairpersons at the **intermediate and district** levels (**243C**).	**Reservation for backward classes (OBCs)**.
Reservation for **SCs and STs** in proportion to population, and **not less than one-third** of seats and chairperson posts for **women** (**243D**).	**Powers, functions and funds** from the **29 Eleventh Schedule** subjects (**243G**); **taxes** panchayats may levy (**243H**).
Five-year term; after dissolution, **fresh election within six months** (**243E**); minimum age **21** to contest (**243F**).	Raising women's reservation **above one-third** (many States, including **AP**, give **50%**).
State Finance Commission every **five years** (**243I**); **State Election Commission** to conduct elections (**243K**).	**Audit** arrangements (**243J**) are made by State law.

- **Exempted areas (Art 243M): Nagaland, Meghalaya, Mizoram**; hill areas of **Manipur** with district councils; **Darjeeling** (Gorkha Hill Council); **Scheduled and tribal areas**. For **Fifth Schedule areas**, Parliament passed the **PESA Act, 1996** (**Dileep Singh Bhuria Committee**), which gives the **Gram Sabha** strong powers over **minor forest produce, land acquisition consultation, minor minerals and local markets**. **Art 243-O** bars courts from interfering in panchayat **electoral matters**.
- **Andhra Pradesh:** the **AP Panchayat Raj Act, 1994** set up three tiers: **Gram Panchayat** (sarpanch **directly elected**), **Mandal Praja Parishad** (MPTC members elected; president chosen **indirectly**) and **Zilla Praja Parishad** (ZPTC members elected; chairperson chosen **indirectly**). The **Panchayat Secretary** is the village-level official, now working from **Village/Ward Secretariats** (since **2019**).
- **Issues and recent steps:** too little **own revenue**, **funds, functions and functionaries** not fully transferred, **'sarpanch-pati'** (husbands acting for elected women), weak **Gram Sabhas**. **OBC quota** must pass the **'triple test'** (**K. Krishna Murthy 2010; Vikas Gawali 2021**). Steps: **Finance Commission grants**, **Gram Panchayat Development Plans**, **e-GramSwaraj**, **SVAMITVA (24 April 2020)** property cards, **Rashtriya Gram Swaraj Abhiyan**.

EXAM TRAP

Balwant Rai Mehta 1957: **3 tiers** (GP, samiti, ZP), samiti **executive**, ZP advisory (**Collector chair**); NDC **Jan 1958**; **Rajasthan (Nagaur) 2 Oct 1959**, **AP 1959**; **Ashok Mehta 1977–78**: **2 tiers** (ZP, mandal panchayat 15–20 thousand), ZP **executive**, **parties**, nyaya panchayats; **G.V.K. Rao 1985** ('grass without roots'); **L.M. Singhvi 1986** (constitutional status); **Thungon 1988**; **64th Bill 1989 failed**; **73rd: passed Dec 1992, in force 24 Apr 1993**; **Part IX, 243–243-O**, **11th Schedule, 29 subjects**; **243A Gram Sabha**; **243B** (skip middle tier below **20 lakh**); **243D** SC/ST, women **≥1/3** (AP **50%**); **243E** 5 years, **6 months**; **243F age 21**; **243I SFC**;

243K SEC; **243M exemptions**; **PESA 1996 (Bhuria)**; **AP PR Act 1994**: GP, MPP, ZPP; triple test; **SVAMITVA 2020**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about the Balwant Rai Mehta Committee:

1. It recommended a three-tier Panchayati Raj system.

2. It wanted the zila parishad to be the main executive body.

3. Rajasthan was the first State to adopt its system, at Nagaur in 1959.

4. Andhra Pradesh adopted Panchayati Raj in 1959.

Which of the above are correct?

A. 1 and 3 only

B. 1, 3 and 4 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: B. Balwant Rai Mehta made the panchayat samiti the executive body; the zila parishad was advisory. The Ashok Mehta Committee made the zila parishad executive.

PRACTICE

Match List-I (committee) with List-II (recommendation):

List-I (committee)	List-II (recommendation)
A. Ashok Mehta	1. Constitutional recognition for panchayats, 1986
B. G.V.K. Rao	2. Three-tier system, 1957
C. L.M. Singhvi	3. Two-tier system with mandal panchayats, 1978
D. Balwant Rai Mehta	4. 'Grass without roots', 1985

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 3 C → 1 D → 2

(3) A → 3 B → 1 C → 4 D → 2

(4) A → 3 B → 4 C → 1 D → 2

Answer: (4) Ashok Mehta: two tiers; G.V.K. Rao: grass without roots; Singhvi: constitutional status; Balwant Rai Mehta: three tiers.

PRACTICE

Which one of the following statements about the 73rd Amendment is NOT correct?

A. A State with a population below 20 lakh need not have panchayats at the village level.

B. Not less than one-third of seats are reserved for women.

C. Panchayat elections are conducted by the State Election Commission.

D. It added the Eleventh Schedule with 29 subjects.

Answer: A. Small States may skip only the intermediate tier, not the village tier.

PRACTICE

Assertion (A): National Panchayati Raj Day is observed on 24 April.
Reason (R): The 73rd Amendment received the President's assent on 24 April 1993.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: C. A is true. R is false: the Act received the President's assent on 20 April 1993 and came into force on 24 April 1993; the day marks its coming into force.

PRACTICE

A State Government dissolves a gram panchayat before its term ends. Under the Constitution, within what period must the new election be completed?

A. Within one year

B. Within three months

C. At the next general panchayat election only

D. Within six months of dissolution

Answer: D. Art 243E requires an election within six months of dissolution.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the Balwant Rai Mehta and Ashok Mehta recommendations side by side. 2. Write the later committees and the 64th Bill. 3. Write the 73rd Amendment: dates, Articles, compulsory and voluntary provisions, exemptions, PESA and AP's system.

SCIENCE · Ecology and ecosystems

UNDERSTAND IT FIRST

What is it? Ecology is the study of how **living things relate to each other and to their surroundings**. An **ecosystem** is a working unit where **plants, animals and microbes (biotic)** interact with **air, water, soil, light and temperature (abiotic)**, for example a pond, a forest or the **Kolleru lake**. In every ecosystem, **energy flows** in one direction (from the Sun through plants to animals) and **nutrients cycle** round and round. Understanding this explains **why tigers are few, why pesticides harm eagles and why lakes turn green**. Learn it as **terms → components → food chains and energy → pyramids → cycles → succession → interactions → key concepts**.

1. Basic terms and components

- The word **'ecology'** was coined by **Ernst Haeckel (1866)**; **'ecosystem'** by **A.G. Tansley (1935)**. **Levels:** individual → **population** (same species in an area) → **community** (all species in an area) → **ecosystem** (community + non-living surroundings) → **biome** (large region with similar climate and vegetation, e.g. tropical rainforest, desert) → **biosphere** (all life on Earth). **Habitat** is the **address** of an organism; **niche** is its **'profession'**, its full role (what it eats, when it is active, what eats it).
- **Biotic components: producers (autotrophs)**, mainly green plants and algae that make food by **photosynthesis**; **consumers (heterotrophs)**: **primary** (herbivores), **secondary** and **tertiary** (carnivores), **omnivores**; **decomposers** (bacteria, fungi) that break dead matter into nutrients. **Abiotic components:** sunlight, temperature, water, soil, air and nutrients.

2. Energy flow, food chains and pyramids

Pyramid of energy and the 10% law

Idea	Explanation
Food chain and food web	A **food chain** shows who eats whom (grass → deer → tiger). The **grazing** chain starts with living plants; the **detritus** chain starts with **dead matter** (important in forests and

	mangroves). Chains interlink to form a **food web**, which makes the ecosystem **stable**. Each step is a **trophic level**.
Energy flow and the 10% law	Energy flow is **one-way (unidirectional)**: it cannot go back to the Sun. **Lindeman's 10% law (1942)**: only about **10%** of energy passes to the next level; the rest is used in living or lost as **heat**. That is why food chains are **short (3–5 levels)** and **top predators are few**.
Ecological pyramids	**Pyramid of numbers**: usually upright, but **inverted in a tree ecosystem** (one tree, many insects). **Pyramid of biomass**: upright on land, but **inverted in the ocean** (small mass of fast-growing phytoplankton supports a larger mass of fish). **Pyramid of energy**: **always upright**.
Productivity	**Gross primary productivity (GPP)** is the total food made by plants; **net (NPP)** = GPP minus what plants use in respiration, the food available to consumers. **Tropical rainforests**, **estuaries, coral reefs and swamps** are the most productive; **deserts** and the **open ocean** (per area) the least.

3. Nutrient cycles and succession

- **Biogeochemical cycles** move nutrients between living things and the environment. **Carbon cycle**: photosynthesis takes CO_2 in; respiration, decay and **burning fossil fuels** release it. **Nitrogen cycle**: **fixation** of air nitrogen by **Rhizobium** (in root nodules of **legumes**), **Azotobacter** and **cyanobacteria** (Anabaena, Nostoc) and by **lightning**; **nitrification** by **Nitrosomonas** (ammonia → nitrite) and **Nitrobacter** (nitrite → nitrate); **denitrification** by **Pseudomonas** returns N_2 to the air. **Phosphorus cycle** is **sedimentary** (no gas phase; source is **rocks**).
- **Ecological succession** is the slow, orderly change of communities in an area until a stable **climax community** forms. **Primary succession** starts on **bare rock or new land** (lava, sand dunes) where there was no soil: **lichens** are the **pioneers**, then mosses, grasses, shrubs and trees. **Secondary succession** happens where life was removed but **soil remains** (after fire, flood, an abandoned field) and is **faster**. Succession in water is a **hydrosere**; on dry rock, a **xerosere**.

4. Interactions and key concepts

Term	Meaning and example
Mutualism (+/+)	Both gain: **lichen** (alga + fungus), **mycorrhiza** (fungus + plant roots), Rhizobium in legumes, bees and flowers.
Commensalism (+/0)	One gains, the other is unaffected: an **orchid** growing on a mango branch; **cattle egret** and grazing cattle.
Parasitism and predation (+/−)	**Cuscuta** (dodder) on a host plant; lice, tapeworm; a tiger hunting deer.
Competition (−/−) and amensalism (−/0)	Species compete for the same resource; in amensalism one is harmed and the other is unaffected, e.g. **Penicillium** mould stopping bacteria.
Keystone and flagship species	A **keystone species** has an effect far bigger than its numbers; removing it changes the whole ecosystem (e.g. **tiger**, **fig trees**). A **flagship species** is a popular animal used to

	win support for conservation (tiger, elephant).
Ecotone and edge effect	An **ecotone** is a **transition zone** between two ecosystems (e.g. **mangroves** between land and sea, a forest–grassland border); it often has **more species** (**edge effect**).
Biomagnification and eutrophication	**Biomagnification**: non-degradable poisons (**DDT, mercury**) become **more concentrated at each higher level**; fish-eating birds lay **thin-shelled eggs**. **Eutrophication**: **nitrates and phosphates** from fertilisers and sewage cause **algal blooms**; when algae die and decay, **oxygen falls** and fish die.

Ecosystems in Andhra Pradesh: Eastern Ghats forests (**Araku and Paderu** hills in the north, **Nallamala** and **Seshachalam** in the south); **Kolleru** (a large **freshwater lake** between the Godavari and Krishna deltas, a **Ramsar site**); **Pulicat** (India's **second-largest brackish lagoon**); **mangroves** at **Coringa** (Godavari) and the **Krishna** estuary; **grasslands** of **Rollapadu** (home of the Great Indian Bustard).

EXAM TRAP

Haeckel 1866 (ecology); **Tansley 1935** (ecosystem); population → community → ecosystem → biome → biosphere; habitat = address, niche = role; producers, consumers, decomposers; **grazing** and **detritus** chains; food web; **energy one-way**; **Lindeman 10% (1942)**; pyramids: numbers **inverted (tree)**, biomass **inverted (ocean)**, energy **always upright**; **GPP, NPP**; rainforests, estuaries, reefs most productive; N-fixers **Rhizobium, Azotobacter, Anabaena, Nostoc**, lightning; **Nitrosomonas, Nitrobacter**; denitrifier **Pseudomonas**; **phosphorus sedimentary**; succession: **lichens pioneer**, climax; primary vs secondary; hydrosere, xerosere; **mutualism (lichen, mycorrhiza)**, **commensalism (orchid)**, **parasitism (Cuscuta)**, **amensalism (Penicillium)**; keystone, flagship; **ecotone, edge effect**; **biomagnification (DDT)**; **eutrophication (N, P)**; AP: Eastern Ghats, **Kolleru, Pulicat, Coringa, Rollapadu**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. Energy flow in an ecosystem is unidirectional.

2. The pyramid of energy is always upright.

3. The pyramid of biomass in an ocean is usually inverted.

4. Decomposers occupy the first trophic level.

Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2, 3 and 4

D. 1, 2 and 3 only

Answer: D. Producers (green plants) occupy the first trophic level; decomposers work at all levels.

PRACTICE

Match List-I (interaction) with List-II (example):

List-I (interaction)	List-II (example)
A. Mutualism	1. Penicillium stopping the growth of bacteria
B. Commensalism	2. Cuscuta on a host plant
C. Parasitism	3. An orchid on a mango branch
D. Amensalism	4. Lichen

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 2 C → 3 D → 1

(3) A → 4 B → 3 C → 2 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (3) Lichen: mutualism; orchid: commensalism; Cuscuta: parasitism; Penicillium: amensalism.

PRACTICE

Which one of the following pairs is NOT correctly matched?

A. Ernst Haeckel : coined the term 'ecology'

B. Rhizobium : denitrification

C. A.G. Tansley : coined the term 'ecosystem'

D. Lindeman : ten per cent law

Answer: B. Rhizobium fixes nitrogen in legume root nodules; Pseudomonas carries out denitrification.

PRACTICE

Assertion (A): The concentration of DDT is highest in fish-eating birds in an aquatic food chain.
Reason (R): Non-biodegradable chemicals become more concentrated at each higher trophic level.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. This is biomagnification, which explains A.

PRACTICE

A volcanic eruption leaves a fresh layer of bare lava rock with no soil. Which organisms are most likely to colonise it first?

A. Grasses

B. Shrubs

C. Lichens

D. Large trees

Answer: C. In primary succession on bare rock, lichens are the pioneers; they break rock and start soil formation.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the terms, levels and components. 2. Write food chains, the 10% law, the three pyramids and productivity. 3. Write the nitrogen cycle with organisms, succession, interactions with examples and the key concepts, with AP's ecosystems.

NOTEBOOK PAGE · Day 25: write this from memory

Close the notes. Write these headings, fill every blank, then correct in a different colour.

GEOGRAPHY: The Earth in the Solar System

Area	Write
Universe and Sun	Big Bang, galaxy, light year, AU; Sun's facts and layers
Planets and small bodies	Order; one fact per planet; dwarf planets; asteroids, comets, meteors
Earth and Moon	Shape and size; rotation, revolution, seasons with dates; latitudes; longitude and time; IST; Moon and eclipses

POLITY: Panchayati Raj

- Balwant Rai Mehta (year, tiers, key points, first States); Ashok Mehta (tiers, points); later committees; 73rd Amendment: Articles, Gram Sabha, tiers, elections, reservation, term, SEC, SFC, Eleventh Schedule; exemptions and PESA; AP's structure.

SCIENCE: Ecology and ecosystems

Write: terms and who coined them; levels of organisation; components; food chain, web, trophic levels; 10% law; three pyramids; productivity; cycles; succession; interactions with examples; key concepts (keystone, ecotone, biomagnification, eutrophication).